

NETWORK PROBLEM 6:

IoT NETWORK SECURITY ISSUES (LAYER 1-7 – MULTI-LAYER THREATS)

IoT (Internet of Things) Network Security Issues refer to vulnerabilities and risks associated with the growing number of interconnected smart devices. These devices often lack proper security controls, making them prime targets for attackers. Since IoT devices operate across multiple layers of the OSI model, their weaknesses can result in data theft, unauthorized access, and full-scale network compromise.

Causes of IoT Network Security Issues:

1. Weak or Default Credentials:

- Many IoT devices are shipped with factory-set usernames and passwords. ○ Users often fail to change these, allowing easy access for attackers using brute force or credential stuffing techniques.

2. Lack of Firmware Updates:

- IoT devices frequently run outdated software. ○ Manufacturers may not provide updates, or users may neglect to apply them, leaving known vulnerabilities unpatched.

3. Unencrypted Communication:

- Data transmitted by IoT devices may not use secure protocols (e.g., HTTPS, TLS). ○ This makes it easier for attackers to intercept and manipulate data using packet sniffers or MITM attacks.

4. Insecure Network Services:

- Unnecessary open ports or services (e.g., Telnet, FTP) on devices increase the attack surface.
- Attackers exploit these services to gain unauthorized access.

5. Insufficient Authentication and Authorization:

- Devices may lack proper user authentication or role-based access control.
- This allows unauthorized users to perform sensitive operations.

6. Lack of Device Monitoring and Logging:

- Without proper logging, it's difficult to detect suspicious activity or track incidents.

7. Large Attack Surface:

- The more devices connected to a network, the more entry points exist for an attacker.

8. Physical Access Vulnerability:

- IoT devices often operate in uncontrolled or public environments. Attackers can tamper with the hardware to extract data or modify firmware.

Symptoms of IoT Network Security Issues:

1. Unauthorized Access to Devices:

- Devices performing unexpected operations or configuration changes.

2. Increased Network Traffic or Anomalies:

- Sudden spikes in traffic from IoT devices, especially outbound traffic, can indicate botnet activity (e.g., DDoS attacks).

3. Device Malfunctions or Downtime:

- Infected or hijacked devices may crash, freeze, or stop functioning normally.

4. Suspicious External Connections:

- IoT devices attempting to communicate with unknown or blacklisted IP addresses.

5. Data Leaks:

- Sensitive information (e.g., video feeds, sensor data) being accessed by unauthorized users or publicly exposed.

Resolving IoT Network Security Issues:

1. Change Default Credentials:

- Immediately replace factory-set usernames and passwords with strong, unique credentials on all IoT devices.

2. Apply Regular Firmware Updates:

- Check for and install security patches from manufacturers.
- Consider disabling or replacing unsupported devices.

3. Isolate IoT Devices Using VLANs or Subnets:

- Segment the network to limit IoT devices' access to critical systems and data.

4. Enable Encryption:

- Use secure protocols like HTTPS, TLS, and WPA3 for wireless communication.

5. Disable Unused Services and Ports:

- Close or block unnecessary network services to reduce the attack surface.

6. Implement Network Access Control (NAC):

- Only allow authenticated and authorized devices to connect to the network.

7. Monitor Device Activity:

- Use intrusion detection systems (IDS), firewalls, and monitoring tools to track and alert unusual behavior.

8. Restrict Physical Access:

- Install devices in secure locations or tamper-resistant enclosures.

9. Use Secure IoT Platforms and Vendors:

- Choose manufacturers that follow strong security practices and provide long-term support.

10. Educate Users and Administrators:

- Provide training on IoT risks and safe configuration practices.

Simulation of problem and solution:

Large Attack Surface & VLAN Segmentation:

A large number of IoT devices were deployed and VLANs were used as a solution and the devices were segmented.

